**University of Macedonia**

Computer Science & Applied Informatics, Information Systems

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# The Evolution of Capitalism

## *A **Bittensor** Thesis*

*(Decentralized AI)*

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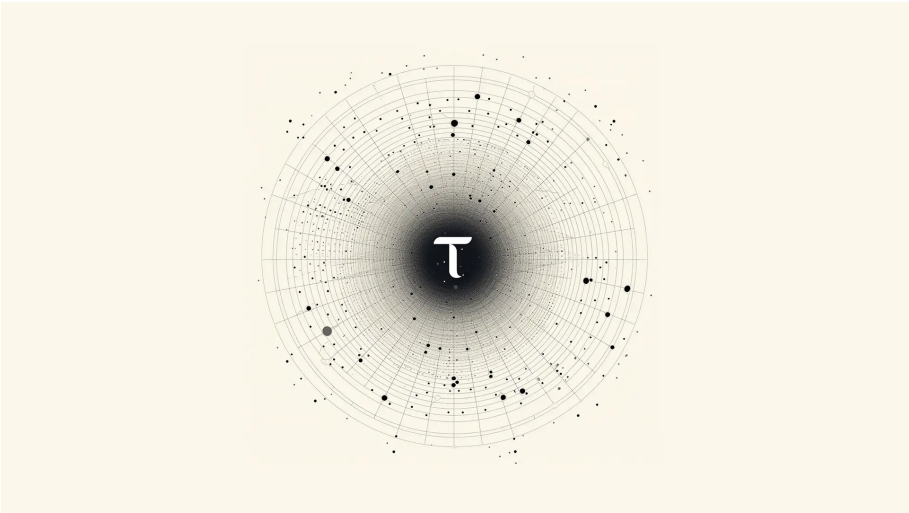

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# Abstract

This thesis explores the possible evolution of capitalism through the lens of decentralised artificial intelligence, with a focus on **Bittensor** ($\tau$)—a Decentralized peer-to-peer protocol that incentivises The Global Intelligence-Models/Code to compete and collaborate in an open, permissionless network, all striving to provide the best **Digital Commodity**.

Beginning with a historical overview of capitalism's major phases—from mercantilism and industrialism to financialised platform economies—the thesis examines how each era redefined value, labour, and ownership. It then investigates how Bittensor's architecture, consensus mechanism, and **Mining Implementation Mechanisms/Architecture** may constitute a structurally new phase in this evolution: one where intelligence itself becomes the commodity, and decentralised networks replace centralised institutions as the primary engines of value creation.

**Keywords:** Bittensor, Decentralised AI, Capitalism, Evolution, Incentives, Yuma Consensus, Mining, Validating, Subnets, Tokenomics.

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# Background and Motivation

The motivation for this thesis did not originate in academic literature. It originated in observation—of systems, of incentives, and of the recurring patterns that emerge when human institutions fail to reward the people who drive them forward.

The first concern was concentration. As technology companies scaled, it became increasingly clear that the entities controlling the most valuable digital commodities—code, data, intelligence—had little structural incentive to share them, improve them, or relinquish control over them. Monopolisation, in this context, is not merely an economic outcome; it is the rational end-state of any system where ownership is centralised and competition is suppressed.

The counterexample came from **GitHub**. By making code openly accessible and attributable, GitHub demonstrated that transparency and shared ownership could unlock contributions at a scale no single company could replicate. Developers contributed not because they were told to, but because the system rewarded effort with visibility, reputation, and influence. Yet even this system operated within a broader economy defined by inflation, wage stagnation, and the persistent misalignment between contribution and compensation.

This led to **Bitcoin**—a protocol that replaced institutional trust with mathematical proof, and rewarded participants directly and transparently for the work they performed in maintaining the network. For the first time, an economic system existed in which the rules could not be arbitrarily changed by those in power providing true Digital Ownership.

Parallel observations emerged from platforms such as **Kaggle**, where data scientists compete openly to solve real problems, ranked by performance and rewarded accordingly. The model was simple but powerful: define a useful task, open it to everyone, reward the best result. The output was not merely a winning model—it

was a global, self-organising meritocracy of intelligence.

The sharpest contrast, however, came from direct experience within conventional corporate structures. The defining feature of the modern firm is not efficiency—it is misalignment. Workers are rarely compensated in proportion to the value they generate. Incentive structures reward tenure over talent, compliance over creativity. Entire departments exist not to produce value, but to manage the friction created by the absence of genuine incentives. Human Resources, in many organisations, functions less as a mechanism for unlocking human potential and more as a bureaucratic bottleneck—a layer of administration interposed between ability and reward.

It was against this backdrop that **Bittensor** ($\tau$) came into focus: a protocol that takes the open competition of Kaggle, the trustless reward mechanism of Bitcoin, and the collaborative transparency of GitHub—and applies them to the production of Intelligence itself. On Bittensor, Models/Participants compete in open subnets, validators score their outputs, and emissions flow directly to those whose intelligence is deemed most useful by the network. There are no intermediaries, no HR departments, no opaque performance reviews. There is only contribution—and its verifiable reward. All Participants are Incentivized to Be Honest and Produce the best Digital Commodity they can produce...

This thesis is an attempt to understand what that means for the future of capitalism itself.

# Chapter 1

# The History and Rise of Centralized Money

The history of money begins with simple systems of exchange, where goods and services were traded directly through barter. As societies expanded and economic activity became more complex, the need for a standardized medium of exchange led to the creation of early forms of money—objects such as grain, metal pieces, and later, minted coins. These forms of money held value because communities collectively agreed upon their worth, allowing trade to occur more efficiently and reliably.

Over time, the authority to issue money shifted from individuals and merchants to centralized institutions. Ancient kingdoms and empires began minting coins stamped with the symbols of rulers, marking the emergence of money as a tool of state legitimacy and control. This process intensified with the development of banking systems, where early banks issued notes, safeguarded deposits, and facilitated large-scale transactions. As trust in these institutions grew, so did their influence over economic life.

The rise of centralized money power accelerated with the creation of national banks and the establishment of government-backed currencies. Paper money, once a substitute for physical commodities like gold or silver, became the dominant form of currency as states asserted control over its issuance and regulation. This transition positioned central banks as key actors in economic management, with the ability to influence interest rates, credit availability, and overall financial stability.

Through these developments, money evolved from a decentralized social agreement into a highly centralized mechanism governed by political and financial authorities. Centralized control over currency provided states with powerful tools for taxation, military funding, economic intervention, and social governance. By concentrating monetary authority, governments and financial institutions gained unprecedented influence over economic systems—marking a fundamental shift in how value, power, and economic activity were organized.

⋆

**Barter Trade → Commodity Money → Metal Coins → Gold Standard → Paper Notes → Fiat Currency → Digital Money & Commodities → (?)**

⋆

# Chapter 2

# The Web3 Era

## 2.1 Decentralized Money

The emergence of decentralized money began with the introduction of **Bitcoin** in 2009, proposed by the pseudonymous **Satoshi Nakamoto**. Bitcoin was a direct response to the vulnerabilities exposed by the **2008 Financial Crisis**—a moment that shattered public trust in banks, governments, and financial intermediaries. Its goal was simple: money that required no trusted third party, operated transparently, and could not be controlled or manipulated by any single entity.

Bitcoin achieves this through a **distributed ledger**—the blockchain—where thousands of independent nodes collectively verify transactions using cryptographic proofs, without any central authority. The **Proof-of-Work** consensus mechanism drives this process: miners compete to solve mathematical puzzles, and the winner earns the right to add the next block and receive newly issued Bitcoin. Security and monetary policy are thus encoded directly into software.

Bitcoin's supply is **capped at 21 million coins**—a fixed limit that stands in deliberate contrast to fiat currencies, which can be printed without bound. This enforced scarcity, combined with its decentralized architecture, established Bitcoin as *digital gold*: a hedge against inflation and monetary manipulation, and the first asset in history to offer true Digitally Economic self-custody at a global scale.

Blockchain technology is widely misunderstood. The core idea—a peer-to-peer

distributed network maintained by its own participants—did not originate with Bitcoin. Its earliest recognizable form was **torrenting**: a decentralized file-sharing protocol through which users simultaneously downloaded and contributed content, with no central server and no single point of control. People built and maintained this network not by mandate, but because the system *aligned their incentives* to do so.

Bitcoin took that same architecture and applied it to money. Just as a torrent node contributes bandwidth in exchange for access, a Bitcoin miner contributes computation in exchange for reward. The insight was not merely technical—it was economic. **Aligned incentives produce genuine innovation.** Bitcoin gave the world the first *deflationary, decentralized, and truly self-custodied* hard money—not because institutions mandated it, but because the protocol made participation individually rational and collectively beneficial.

## 2.2 Altcoins

Despite Bitcoin's revolutionary design, it was never optimally suited to replace fiat, Visa, or banking systems entirely. As stated in Nakamoto's original whitepaper, Bitcoin was conceived as a **peer-to-peer electronic cash system**—yet its prioritisation of security and decentralization came at the cost of transaction throughput. Its slow **TPS (Transactions Per Second)** made it poorly equipped, on its own, to serve the demands of global everyday commerce.

The **Web3 era** represents the expansion of blockchain technology beyond digital money into programmable, decentralized applications. Early projects such as Ethereum, Litecoin, Filecoin etc explored variations in speed, supply mechanics, and use cases. Later networks—Cardano, Polkadot, Solana, Kaspa—introduced scalable architectures and new economic models, shifting the focus from simple value transfer to broader decentralized ecosystems.

Among all these developments, **Ethereum** played the most pivotal role. Launched in 2015, Ethereum introduced **smart contracts**—self-executing programs stored on the blockchain—transforming blockchains from static ledgers into fully expressive global computation platforms. This enabled decentralized applications (dApps), DeFi, NFTs, and thousands of programmable economic systems. Ethereum's later

transition from **Proof-of-Work to Proof-of-Stake** significantly reduced energy consumption, increased TPS, and laid the foundation for further scaling upgrades such as sharding. It remains the dominant smart-contract platform and central building block of Web3.

As the ecosystem matured, distinct networks emerged—each making deliberate trade-offs between speed, security, decentralization, and privacy.

**Ethereum** (PoS, Smart-Contract Pioneer) prioritizes decentralization, security, and developer adoption. Layer-2 rollups now handle most transaction throughput, positioning Ethereum as the settlement and coordination layer of Web3 rather than a high-TPS execution chain.

**Solana** (PoS + Proof of History) is designed for maximum throughput and low latency. Its unique Proof of History mechanism enables very high TPS and low fees, making it well-suited for real-time applications—DeFi, NFTs, and on-chain gaming—at the cost of higher hardware requirements.

**Kaspa** (PoW, blockDAG Architecture) reimagines Proof-of-Work by replacing the traditional linear chain with a **blockDAG (GHOSTDAG)** structure, allowing multiple blocks to be produced and confirmed in parallel. This enables approximately one-second block times while preserving PoW security and decentralization—widely considered a serious attempt at solving the *Blockchain Trilemma* of speed, scalability, and decentralization.

**Sui** (PoS, Object-Based Execution) enables parallel transaction execution through an object-based data model, achieving high throughput and low confirmation times—optimized for games and consumer-facing applications.

**Monero** (PoW, Privacy-First) uses ring signatures, stealth addresses, and confidential transactions to ensure that senders, receivers, and amounts remain private by default. It is the only major cryptocurrency used natively for private peer-to-peer transactions, including on dark web markets—a distinction that underscores both its utility and its controversy.

**Zcash** (PoW, Selective Privacy) introduced **zk-SNARKs**—zero-knowledge proofs—as the first blockchain to implement them correctly, allowing users to optionally shield transactions, hiding sender, receiver, and amount mathematically, while retaining the option of full transparency.

**Hyperliquid** (HyperBFT, On-Chain Perpetuals) is a Layer 1 blockchain built entirely around a single purpose: a fully on-chain central limit order book delivering sub-second finality at over 200,000 orders per second. No venture capital, no centralised matching engine—by 2024 it held over 55% of the global perpetual DEX market, proving that professional-grade financial infrastructure no longer needs to be centralised.

## 2.3 Consensus Mechanisms

A consensus mechanism is the process a blockchain uses to agree on a single, shared version of the ledger across thousands of independent nodes. It determines who gets to add new blocks, which transactions are valid, and how the network resolves conflicts—all without relying on a central authority.

In simple terms, Consensus is:

*"How do strangers agree on the truth of the Network without trusting each other?"*

Each network makes deliberate trade-offs between security, decentralization, scalability, energy efficiency, and complexity. The following Table shows some of the most significant consensus designs in the current ecosystem.

| Network | Mechanism | Strengths | Weaknesses |
|---|---|---|---|
| **Bitcoin** | Proof of Work | Very high security & decentralization | Low TPS, high energy use, slow finality |
| **Ethereum** | Proof of Stake | Energy efficient, faster finality, scalable | Capital concentration risk, complex protocol |
| **Cardano** | PoS – Ouroboros | Strong security assumptions, sustainable design | Slower development, lower throughput |
| **Solana** | PoS + Proof of History | Very high throughput, low latency | High hardware requirements, centralization pressure |
| **Sui** | PoS – Parallel Execution | High throughput, fast confirmations | Complex execution model, high validator requirements |
| **Kaspa** | PoW – BlockDAG (GhostDAG) | Fast confirmations with PoW security, fair mining | Complex consensus logic, less battle-tested |
| **Monero** | PoW – Privacy Focused | Strong privacy, ASIC-resistant & fair mining | Lower scalability, heavier transaction data |
| **Zcash** | PoW + zk-SNARKs | Optional cryptographic privacy, transparency | High complexity, trusted setup assumptions |
| **Hyperliquid** | HyperBFT – On-Chain CLOB | Sub-second finality, 200k orders/sec, no VC | Single-purpose chain, validator centralisation risk |
| **Bittensor** | Yuma Consensus – Incentive-Based | Aligns incentives with real-world utility, energy efficient | Complex incentive design, harder security reasoning |

Table 2.1: Consensus mechanism trade-offs across major blockchain networks.

**Summary.** No single consensus mechanism optimises all variables simultaneously. Every network makes a deliberate choice about what to prioritise— security, speed,

privacy, or utility—and accepts the corresponding trade-offs. Bittensor's contribution is to extend this design space beyond block production into the domain of *intelligence itself*: consensus not over transactions, but over the Value and Validity of Intelligence.

# Chapter 3

# Bittensor ($\tau$)

## 3.1 Decentralized Intelligence (Web4)

The natural and most logical path in the evolution of digital technologies is the convergence of decentralized systems with the most valuable and promising advancement of our lifetime: Artificial Intelligence. As blockchain matured—from decentralized money to programmable networks—integrating AI into these systems became an inevitable next step. The idea of combining distributed incentives, open participation, and global coordination with intelligent computation attracted significant interest across the industry.

Many Blockchains attempted to merge with AI, exploring visions of decentralized model training, data marketplaces, and AI governance. However, most struggled with scalability, economic viability, or accurately describign what intelligence or model should be valued. Among these efforts, **Bittensor** ($\tau$) emerged as the most coherent and promising implementation— successfully aligning blockchain incentives with the production of high-quality machine intelligence. It built an **Incentive Layer**: a permissionless system in which anyone with the right knowledge and expertise can participate, contribute, and be rewarded— entirely bypassing centralised means of gatekeeping such as location-based hiring, slow institutional processes, and opaque performance evaluation.

At its core, Bittensor represents what can be described as **Web4**: the era of decentralized intelligence. Participants contribute machine learning models—or whatever computational solution the network demands—validators assess their outputs, and the network issues $\tau$ tokens based on performance. Intelligence itself becomes

an *economic resource and* ***commodity***.

This architecture eliminates the need for a centralised AI provider. Instead, the network becomes a simultaneously **competitive and collaborative** marketplace of models, each improving through continuous incentive-driven evolution. Bittensor's design mirrors *natural selection*: high-performing intelligence is rewarded, weaker models fall behind, and the system grows more capable over time.

By enabling anyone in the world to contribute or access intelligence, Bittensor establishes intelligence as a **decentralized public good**. It extends the logic of Bitcoin and Ethereum into a new domain—where not only value or computation is decentralized, but the *creation of intelligence itself*. Bittensor was never solely intended for machine learning goods. It was designed for any problem that can be rigorously expressed and assessed:

*"If you can express your problem and the assessment well enough,*
*the network will solve the problem for you."*

## 3.2 The Network

### 3.2.1 Architecture and Participants

Bittensor's network is not a single monolithic blockchain. It is a **layered ecosystem** of specialised sub-networks, each dedicated to a specific problem or domain of intelligence. At the top level sits the **root network**, which governs the allocation of $\tau$ emissions across the entire ecosystem. Beneath it operate hundreds of independent **Subnets**—each one a self-contained competitive market for a specific type of intelligence or digital commodity. Bittensor is Inflationary by design especially in its early days so it can fund its own Subnets and pay the participants through Emissions.

The network is composed of four core participants:

**Subnet Owners** are the architects of their own Sub-System(Subnet). They define the problem, design the incentive mechanism, and deploy the subnet. In economic terms, they function as *entrepreneurs-Company*—identifying a valuable problem, structuring the rules of competition, and attracting participants to

solve it. A subnet owner's long-term reward is tied directly to the quality and utility of their subnet.

**Miners** are the producers of intelligence. They run computational nodes—models, algorithms, or any software capable of responding to the subnet's task—and compete to provide the best possible output(Based on the Subnet's Incentive Mechanism-Optimizer). Miners are rewarded in $\tau$ proportional to the quality of their contributions as assessed by validators. Critically, there is no fixed salary, no tenure, and no bureaucratic review—only verifiable performance.

**Validators** are the quality arbiters of the network. They query miners, evaluate their outputs, and assign score Matrixes that determine the distribution of emissions. Validators stake $\tau$ as collateral, aligning their economic incentives with honest and accurate assessment. A validator that scores dishonestly or carelessly risks losing both reputation and stake.

**Stakers (Delegators)** are the capital layer of the network. Rather than running nodes or evaluating outputs directly, stakers delegate their $\tau$ to validators they trust, earning a proportional share of the validator's emissions in return. Staking serves two functions simultaneously: it *secures the network* by increasing the economic cost of dishonest validation, and it *democratises participation*—allowing anyone who holds $\tau$ to contribute to and benefit from the network's growth without requiring technical expertise. In economic terms, stakers function as *passive capital allocators*, voting with their stake on which validators—and by extension, which subnets—deserve to grow.

### 3.2.2 Subnets as Companies, Miners as Workers

The most intuitive analogy for understanding Bittensor's structure is that of a **decentralized labour market**. Each subnet functions like an independent company: it has a defined purpose, a set of performance standards, and a budget— denominated in $\tau$ emissions—that it distributes to those who contribute most effectively.

Miners operate within this structure as workers—but workers in a fundamentally different sense from the conventional employment model. There is no hiring process as we know it, no geographic restriction, and no fixed compensation. A miner anywhere in the world can join any subnet, deploy a node, and begin earning immediately if their contribution meets the network's standard and thus producing Intelligence. The incentive is direct, transparent, and proportional to output.

This stands in sharp contrast to the centralised firm, where compensation is decoupled from marginal contribution, and where institutional friction—interviews, hierarchies, HR departments—interposes itself between talent and reward. On Bittensor, **the market itself is the employer, and performance is the only credential that matters**.

### 3.2.3 How Bittensor Differs from Traditional Blockchain Architectures

Bittensor's relationship with its participants is fundamentally unlike that of Bitcoin, Ethereum, or any conventional blockchain network. In traditional systems, miners and validators perform a **fixed, well-defined task**: hash a block, validate a transaction, attest to a slot. The work is repetitive, standardised, and does not require constant adaptation over time. A Bitcoin miner running the same hardware and software in 2015 and in 2025 is performing essentially the same operation.

Bittensor inverts this entirely. Because each subnet defines its own task—and because those tasks are competitive and continuously evolving—**miners must constantly improve their models, update their code, and adapt their nodes** to remain competitive. A miner who deploys a model and leaves it unchanged will inevitably be outperformed by those who iterate. The network does not reward existence—it rewards *improvement*. The network naturally Searches for participants who understand the problem deeply enough to keep improving their solution—transforming the act of participation into an ongoing research and development process.

In this sense, Bittensor is less analogous to a blockchain and more analogous to a **living market for intelligence**.

## 3.3 Tokenomics

### 3.3.1 The Native Tokens

Bittensor's economic architecture is anchored by $\tau$, the network's native token. By design, Bittensor adopts Bitcoin's fair-launch principles: no venture capital allocation,

no pre-mine, and a **hard supply cap of 21 million** $\tau$. The network is owned by the people providing value to it, not by a central board of institutional investors. Every $\tau$ in existence was earned through contribution to the network. Subnets also have their own Token called **Alpha**. it has the exact same tokenomics as $\tau$

### 3.3.2 Halvings

#### What is a Halving?

A halving is a pre-programmed event encoded directly into the protocol in which the rate of new token emissions is cut in half. It is not a decision made by any authority—it is a mathematical certainty, triggered automatically when a specific supply threshold is reached(when Supply doubles) . The purpose is twofold: to enforce **scarcity** as adoption grows, and to transition the network from an incentive-heavy bootstrapping phase toward a mature, utility-driven economy. Each halving makes every previously earned token more scarce—and therefore, assuming sustained or growing demand, more valuable.

#### Two Halving Events in Bittensor

Uniquely, Bittensor operates with **two distinct and independent halving mechanisms**—one at the network level and one at the subnet level:

**TAO Halving (Network-Level)** occurs each time TAO Supply doubles. The first TAO halving occurred in **December 2025**, reducing daily emissions from **7,200** $\tau$ to **3,600** $\tau$. Subsequent halvings will continue compressing global emissions over time, mirroring Bitcoin's deflationary monetary trajectory.

**Alpha Token Halving (Subnet-Level)** occurs each time a specific subnet's Supply doubles. It is **independent**—triggered not by a global clock, but by that subnet's own internal supply milestone. This means that older, more established subnets with larger circulating supplies have possibly already halved more times than newer Subents creating a natural *lifecycle* for each subnet's economy.

This dual-halving architecture means that scarcity operates at two levels simultaneously: globally across the entire Bittensor network, and locally within each individual subnet market. The result is a richly layered deflationary system in which both $\tau$ and each subnet's Alpha token are subject to their own independent supply constraints.

### 3.3.3 Emissions

Emissions are the mechanism by which new $\tau$ is introduced into circulation(Inflation). Rather than being minted arbitrarily or distributed by a central authority, emissions are **algorithmically released**(3600 $\tau$ each day currently) by the protocol at a fixed rate and allocated directly to network participants based on their verified contribution.

Every block produced on the Bittensor blockchain generates a fixed quantity of new $\tau$, which flows first to the **root network**—the top-level layer—and is then allocated downward to individual subnets proportional to the amount of $\tau$ staked into each subnet's pool. A subnet that attracts more staked capital receives a larger share of the global emissions. This creates a direct market signal: **capital flows toward subnets producing genuine utility** (just like in our own capitalistic economy), and emissions follow capital.

Within each subnet, received emissions are then distributed among its four productive participants—subnet owners, miners, validators and stakers.

### 3.3.4 Emission Distribution

Emissions flowing into each subnet are then split in half. Half are given to the Subnets's TAO pool and the rest 50% are allocated across participants according to Default protocol-level proportions:

| Participant | Share | Role |
|---|---|---|
| Subnet Owner | 18% | Maintains infrastructure and incentive mechanism |
| Miners | 41% | Produce the computational work and intelligence |
| Validators | 41% | Score miners and distribute rewards via weighted consensus |
| Stakers | - | Stake their $\tau$ with a Validator showing Trust and also receiving $\tau$ from him. |

Table 3.1: TAO emission distribution per subnet.

Stakers earn indirectly through validators—receiving a proportional share of the validator's 41% allocation, minus a validator fee typically set at 18%. Note that those Percentages are set by Default but they can change from the Subnet Owner.

### 3.3.5 Subnet Alpha Tokens ($\alpha$)

With the introduction of **Dynamic TAO** upgrade, each subnet was assigned its own native token—the **Alpha token ($\alpha$)**—also hard-capped at 21 million units. This created a two-layer economic architecture: $\tau$ as the global reserve currency of the network, and Alpha tokens as subnet-level instruments that reflect the specific utility and demand of each subnet's market.

Each subnet operates an internal **Automated Market Maker (AMM)**—analogous to a Uniswap liquidity pool—through which participants exchange $\tau$ for a subnet's Alpha tokens (unlike traditional markets that rely on an order book). The price of a subnet's Alpha token is determined entirely by market forces, governed by the **constant product formula**:

$$k = \tau_{\text{pool}} \times \alpha_{\text{pool}}$$

Where $k$ is a constant, $\tau_{\text{pool}}$ is the amount of $\tau$ in the pool, and $\alpha_{\text{pool}}$ is the amount of Alpha tokens in the pool. The spot price of one Alpha token in terms of $\tau$ is therefore:

$$P_\alpha = \frac{\tau_{\text{pool}}}{\alpha_{\text{pool}}}$$

When a participant stakes $\tau$ into the pool, $\tau_{\text{pool}}$ increases, $\alpha_{\text{pool}}$ decreases, and the price of Alpha rises. When they unstake, the reverse occurs. No counterparty is needed—the math is the market maker.

This mechanism transforms subnet valuation into a *continuous market referendum* on which subnets are producing genuine utility: capital flows in, price rises, emissions follow.

The implications are significant. Capital flows naturally toward high-performing subnets, increasing their Alpha token price and attracting more miners and validators. Underperforming subnets face capital outflows, reduced emissions, and competitive pressure to improve or be replaced. **The market itself becomes the Price regulator**.

## 3.4 Yuma Consensus

Yuma Consensus is the mechanism Bittensor uses to determine **what Miners and Validators get paid and how much**. It is not a block-production consensus like Bitcoin's Proof-of-Work or Ethereum's Proof-of-Stake. Its sole purpose is to produce a **fair, manipulation-resistant distribution of emissions**.

### How It Works

Yuma Consensus runs in five sequential steps each tempo:

**Step 1 — Stake Normalisation**

Each Validators's Stake[1] is normalised so all stakes sum to 1:

$$s_v = \frac{\tau_v}{\sum_v \tau_v}$$

**Step 2 — Rank**

Each miner receives a rank(from Validators): the stake-weighted sum of all validator scores:

$$R_m = \sum_v s_v \cdot W_{v,m}$$

[1] See Glossary: Validator Stake

**Step 3 — Trust**

A miner's trust reflects how consistently validators agree on their quality:

$$T_m = \sum_v s_v \cdot \mathbb{1}[W_{v,m} > 0]$$

A miner scored by more high-stake validators has higher trust.

**Step 4 — Incentive**

A miner's final emission share combines rank and trust:

$$I_m = R_m \cdot T_m$$

**Step 5 — Validator Dividends**

Each validator's reward is proportional to their bond-weighted[2] share of miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

Each step builds on the previous — stake determines rank, rank combined with trust determines incentive, and bonds determine how much of that incentive each validator captures.

### Yuma Outputs

Yuma Consensus produces two distinct outputs every tempo:

**Miners — Incentives** are determined purely by rank and trust:

$$I_m = R_m \cdot T_m$$

A miner cannot influence their bonds—only their output quality, which drives both their rank and the consistency with which validators score them.

**Validators — Dividends** are determined by their bond-weighted share of miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

Bonds reward validators who **independently discover** high-performing miners early. A validator who scores a great miner before everyone else builds a strong

[2] See Glossary: Bond

bond—and captures a larger share of that miner's incentive stream as a result.

Subnet owners and stakers are paid **outside** of Yuma entirely. The owner's 18% is a fixed protocol cut taken before Yuma runs. Stakers earn proportionally from their delegated validator's dividend, minus the validator's fee.

#### Why It Is Manipulation-Resistant

The critical insight of Yuma Consensus is that it **penalises validators who deviate from the consensus**. A validator that assigns weights very differently from the rest of the validator set—whether out of dishonesty, collusion, or negligence—receives reduced emissions themselves. This creates a powerful self-correcting dynamic:

- Honest validators who assess miners accurately are rewarded.
- Validators who attempt to artificially boost specific miners at the expense of others are penalised.
- Collusion requires controlling a majority of stake—making attacks economically expensive.
- Weigth Copying is economically not viable and it is penalized.

Note: Validators Weigth-Copying method is only viable if the Subnet is constantly not improving which means a better subnet will come and dethrone it.

#### In Simple Terms

Yuma Consensus can be understood as a **stake-weighted peer review system**. Validators are the reviewers, miners are the researchers, and $\tau$ emissions are the funding. The network rewards work that the majority of trusted reviewers agree is valuable, punishes anyone who tries to game the review process but also rewards new improvements. The result is a system where **honesty is the most profitable strategy**.

### 3.4.1 Dynamic TAO

Prior to Dynamic TAO, the allocation of emissions across subnets was determined by a small set of root network validators—a centralised bottleneck that created political pressure and misaligned incentives at the top level of the network.

**Dynamic TAO removed this bottleneck.** Subnet emission allocation is now determined by the market itself—specifically, by how much $\tau$ is staked into each subnet's Alpha token pool. More staked $\tau$ means a higher subnet valuation, which means a larger share of global emissions.

In simple terms: **the market votes with capital**. Subnets that produce genuine utility attract stakers, which increases their emissions, which attracts better miners and validators—a self-reinforcing cycle of quality improvement. Subnets that fail to deliver lose capital, lose emissions, and are forced to improve or die.

### TAO Flow: From Protocol to Participant

Bittensor currently emits approximately **0.5 $\tau$ per block**. With a block time of approximately 12 seconds and a **tempo** of 360 blocks, this produces:

| **Period** | **Pre-Halving** | **Post-Halving (Dec 2025)** |
|---|---|---|
| Per Block | 1.0 $\tau$ | 0.5 $\tau$ |
| Per Tempo (360 blocks) | 360 $\tau$ | 180 $\tau$ |
| Per Day (7,200 blocks) | 7,200 $\tau$ | 3,600 $\tau$ |

Table 3.2: TAO emission rates before and after the first halving.

These emissions flow through the network in three distinct stages:

**Stage 1 — Root Network** Every block, the protocol mints new $\tau$ and delivers it to the **root network**. The root network then allocates this $\tau$ across all active subnets that deserve it . Under Dynamic TAO, each subnet's share is determined entirely by how much $\tau$ is staked into its Alpha pool relative to the total staked across all subnets:

$$\text{Subnet Emission Share} = \frac{\text{TAO staked in subnet pool}}{\text{Total TAO staked across all subnets}}$$

A subnet with 10% of total network stake receives 10% of that block's global $\tau$ emissions. Capital determines allocation.

**Stage 2 — Subnet Level** The $\tau$ allocated to a subnet is converted into that subnet's **Alpha tokens** via its internal AMM pool and distributed among the subnet's productive participants according to Yuma Consensus.

**Stage 3 — Individual Participants** Within the miner and validator pools, emissions are not distributed equally—they are distributed **proportionally to performance**. A miner's share of the 41% is determined by their Yuma Consensus score. A validator's share of the 41% is determined by the amount of $\tau$ staked to them. Stakers receive their cut from the validator's allocation, minus the validator's fee.

## 3.5 Subnets

### 3.5.1 What is a Subnet?

A Subnet is a **specialised, self-contained competitive market** built on top of the Bittensor protocol, dedicated to a specific computational task or digital commodity. If Bittensor is the global brain, a subnet is a specific functional lobe—purpose-built, independently governed, and ruthlessly competitive.

Each subnet defines its own problem, its own metric of success, and its own incentive mechanism. Miners compete to solve the problem. Validators judge the quality of solutions. Emissions reward the winners. There is no fixed salary, no guaranteed participation, and no protection from being outperformed.

This architecture deliberately mirrors **Darwinian natural selection**. Every tempo, every miner's output is evaluated and scored. Those who perform best capture the largest share of emissions. Those who fall behind receive less and less until participation becomes economically unviable. The subnet does not care who you are—only what you produce. Models that do not improve are replaced by models that do. Subnets that do not attract capital and talent stagnate and die.

This is not a metaphor. It is the **literal economic structure** of the network: continuous selection pressure, applied every twelve seconds, across every subnet simultaneously. The result is a system that grows more capable over time because the incentive structure makes improvement the only rational long-term strategy.

Note: There are 256 participants Max per Subnet(Currently). Per subnet, by default:

Total UIDs: 256 (all participants combined) Validators: up to 64 slots (top 64 by emissions get validator permits) Miners: up to 192 slots (remaining 256  64). In practice most subnets have only 5–15 active validators — the 64 is just the maximum ceiling. Unused validator slots are automatically available to miners. Also The subnet owner can change these — they can increase or decrease the maximum number of validators via the max_validators hyperparameter.

### 3.5.2 How Subnets Should Work

A well-functioning subnet follows a clear operational logic. The subnet owner defines a **precise, measurable task**—one where good and bad outputs can be objectively distinguished. Validators query miners, score their outputs according to the owner's incentive mechanism, and submit weights to the blockchain. Yuma Consensus aggregates those weights into consensus scores, and emissions flow to miners proportionally.

The quality of a subnet is almost entirely determined by the quality of its **incentive mechanism**. A well-designed mechanism creates a virtuous cycle: clear evaluation criteria attract serious miners, serious miners produce high-quality outputs, high-quality outputs attract stakers, stakers increase the subnet's emission share, and higher emissions attract even better miners. Conversely, a poorly designed mechanism—one with ambiguous scoring, gameable metrics, or misaligned rewards—produces a race to the bottom.

The subnet owner's 18% emission share is therefore not passive income. It is a **continuous incentive to maintain and improve the mechanism**. An owner who neglects their subnet watches its Alpha token price fall, its emission share shrink, and its best miners migrate to more rewarding subnets.

### 3.5.3 Subnet Lifecycle

**Registration.** Any participant can register a new subnet by paying a registration fee denominated in $\tau$. This fee is burned, reducing circulating supply. As demand for subnet slots grows, the registration cost increases dynamically—ensuring that only subnets with genuine economic conviction enter the network.

**Bootstrapping.** A newly registered subnet begins with zero miners, zero validators, and zero staked capital. The subnet owner must attract participants by designing a compelling incentive mechanism and demonstrating the utility of the task. Early emissions are low, but so is competition— creating an opportunity window for early miners to accumulate bonds and establish dominance.

**Growth.** As the subnet proves its utility, stakers allocate $\tau$ to its Alpha pool. Its emission share grows. More miners and validators register. Competition intensifies. The quality of outputs improves. The Alpha token price rises as a market signal of the subnet's value to the broader network.

**Maturity.** A mature subnet operates as a stable, competitive market with established validators, a diverse miner set, and consistent output quality. Its Alpha token has undergone at least one halving, signalling that a significant portion of its supply has been distributed to productive participants.

**Death.** The network is capped at **128 subnets**(Currently). When a new subnet attempts to register and the cap is reached, the protocol automatically identifies the weakest existing subnet using its **EMA (Exponential Moving Average) price**—a continuously updated average of the subnet's Alpha token price over time:

$$\mathrm{EMA}^{(t)} = \alpha \times P^{(t)} + (1 - \alpha) \times \mathrm{EMA}^{(t-1)}$$

The subnet with the **lowest EMA price** that is outside its immunity period is deregistered—replaced by the incoming subnet. New subnets receive a **4-month immunity period**[3] during which they cannot be deregistered regardless of performance, giving them time to bootstrap. Upon deregistration, all Alpha tokens in the subnet's pool are liquidated and the underlying $\tau$ is distributed proportionally to Alpha holders. The subnet owner receives their registration lock cost back, minus the emissions they already received. The market has rendered its verdict—automatically, mathematically, and without appeal.

[3] See Glossary: Immunity Period

### 3.5.4 Notable Subnets

#### Subnet 62 — Ridges (Autonomous Software Engineering)

Developed by **Ridges AI**, Subnet 62 tackles one of the most complex problems in the digital economy: **autonomous software engineering**. Miners deploy fully autonomous coding agents that compete to write, test, and debug real software in response to engineering challenges—evaluated purely on output quality. Ridges is an open, permissionless meritocracy: the best autonomous engineer wins, regardless of who built it. Its effectively using compute from Chutes (Subnet 64), demonstrating how subnets can compose and build upon one another.

#### Subnet 64 — Chutes (Decentralized AI Compute)

Developed by **Rayon Labs**, Chutes is the **largest subnet on Bittensor by market capitalisation** and one of the clearest demonstrations of product-market fit in the entire ecosystem. It operates as a decentralised, serverless AI compute platform—allowing developers to deploy and run any open-source AI model via a standard API in seconds, with no infrastructure management required.

Chutes is a direct and economically superior alternative to centralised cloud AI providers, delivering inference at approximately **85% lower cost than AWS** for comparable tasks. As of 2025, Chutes ranked as the **number one inference provider by usage on OpenRouter**—the world's most widely used AI model aggregation platform—placing it ahead of most centralised providers. Miners are GPU owners who earn $\tau$ for serving models reliably and efficiently; the better their performance, the greater their emission share.

#### Subnet 3 — Templar (Decentralized AI Training)

Templar's mission is to become the world's best **decentralised model training platform**: miners contribute GPU compute and gradient updates toward the continuous training of large-scale models, rewarded based on their contribution to reducing model loss.

In 2025, Templar completed the fully decentralised training of **Covenant-72B**—a 72 billion parameter model trained across globally distributed GPUs with no central server or datacentre—operating at $9 \times 10^{17}$ FLOP/s, making it the largest active

decentralised training network ever recorded. **Jack Clark**, co-founder of Anthropic, cited Templar in *Import AI #439* as proof that incentive-driven, permissionless model training at scale is not only theoretically possible—but already happening.[4]

### Subnet 44 — Score (Decentralized Sports Intelligence)

**Score** is a decentralised computer vision platform built on Bittensor, dedicated to collecting, annotating, and analysing football data at scale. Miners receive live video streams of football matches and compete to track player movements, ball positions, and game events in real time using object detection and keypoint analysis—producing structured, machine-readable match data evaluated by validators for accuracy and speed.

The problem Score solves is a real one: a single professional football match currently requires hundreds of hours of manual annotation at a cost of thousands of dollars. Score aims to reduce this by **10x to 100x** while dramatically improving speed and accuracy—making the kind of high-quality analytics previously reserved for elite clubs accessible to any team, scout, or analyst in the world. With partnerships spanning **283 leagues and 400,000 matches** of footage, and a fantasy sports application planned for the **2026 World Cup**, Score demonstrates that Bittensor's incentive architecture is not limited to language models—it can be applied to any domain where quality, speed, and scale of data processing matters.

### 3.5.5 Configurable Subnet Variables

One of Bittensor's most powerful design features is that subnet owners are not locked into a fixed architecture. A range of **hyperparameters** can be adjusted by the subnet owner to tune the competitive dynamics of their subnet:

[4] Jack Clark, *Import AI #439*, January 2026. https://importai.substack.com/p/import-ai-439-ai-kernels-decentralized

| Parameter | Description |
|---|---|
| `tempo` | Number of blocks per scoring round. Controls how frequently emissions are distributed and weights are updated. |
| `max_n` | Maximum number of miners permitted in the subnet. Controls competition density. |
| `min_allowed_weights` | Minimum weight a validator must assign. Prevents validators from zeroing out miners entirely. |
| `max_weight_limit` | Maximum weight a validator can assign to a single miner. Prevents winner-takes-all concentration. |
| `immunity_period` | Number of blocks a new miner is protected from deregistration. Gives new entrants time to establish performance. |
| `bonds_moving_avg` | The decay rate $\alpha$ in the bond update formula. Controls how quickly validators build or lose bonds in miners. |
| `owner_cut` | The subnet owner's share of emissions. Protocol default is 18%. |
| `validator_permit_threshold` | Minimum stake required to act as a validator with scoring weight. Default is 1,000 $\tau$. |

Table 3.3: Configurable subnet hyperparameters.

These parameters give subnet owners precise control over the competitive dynamics of their market—balancing openness to new entrants, protection against gaming, and the speed at which the network adapts to new information. Getting these parameters right is as much an art as a science, and represents one of the most active areas of subnet research and iteration in the ecosystem.

## 3.6 Governance

Bittensor separates two distinct layers of operation that are often conflated: **network participation** and **protocol governance**. Mining, validating, staking, and subnet operation are all governed automatically by the protocol—by mathematics, consensus, and market forces. No human decision is required for emissions to flow, miners to be scored, or subnets to compete. The network runs itself.

Protocol governance—changes to the underlying rules of the network itself, such as hyperparameter updates, subnet registration rules, or core upgrades— operates through a separate **bicameral legislature** composed of two bodies:

**The Triumvirate** consists of three members from the **Opentensor Foundation**—the core development team behind Bittensor. The Triumvirate is responsible for drafting and submitting proposals for protocol changes. It cannot act unilaterally—every proposal requires Senate approval before execution.

**The Senate** is composed of up to the top 12 delegate hotkeys by stake weight—community-elected validators who hold sufficient TAO delegated to them to qualify. Senators vote on proposals submitted by the Triumvirate. A proposal passes only when it receives a majority of Senate approvals ($> 50\%$), after which a Triumvirate member executes it on-chain.

This structure ensures that no single party can unilaterally change the protocol. The foundation can propose but not execute without community approval. The Senate can block but not propose. **Both chambers must agree.**

This governance model is explicitly designed as a **transitional architecture**. As the network matures, the goal is to phase out the Triumvirate entirely and transition to fully community-driven governance— where any TAO holder can propose changes and the entire network votes through its delegated stake weight. Governance, like everything else on Bittensor, is on a trajectory toward full decentralisation.

Everything discussed so far in this thesis—mining, validating, staking, subnet competition, Yuma Consensus, emissions—belongs to the **production layer**: the part of the network responsible for generating, evaluating, and rewarding digital commodities. This layer runs autonomously, every block, without any human intervention.

Governance is an entirely separate concern. It is not about who gets paid or how intelligence is produced—it is about **what rules the network itself runs on**. Specifically, governance controls changes to the **runtime**—the core logic encoded into the Bittensor blockchain (Subtensor) that defines everything from emission schedules and subnet registration costs to consensus parameters and hyperparameter bounds. A runtime upgrade is the blockchain equivalent of changing the constitution: it alters the fundamental rules that every participant—miners, validators, subnet owners, and stakers—operates under.

## 3.7 Considerations

No decentralised network is immune to adversarial behaviour. Bittensor anticipates the rational self-interest of every participant and encodes solutions directly into its economic architecture.

### Part I — Addressed by Protocol Design

#### Weight Copying

A lazy validator copies another's weights instead of independently evaluating miners—receiving emissions without doing real work. If unchecked, the entire validator layer converges on a single set of weights, centralising judgement.

**Solution: Bonds.** Each validator accumulates a bond in every miner they score:

$$B_{v,m}^{(t)} = (1 - \alpha) \cdot B_{v,m}^{(t-1)} + \alpha \cdot W_{v,m}^{(t)}$$

A validator's dividend is proportional to their bond-weighted miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

A copier always lags—building no independent bonds. The validator who discovers a high-performing miner first earns the most. **Independent discovery is the most profitable strategy.**

#### Miner Feedback Gaming

A miner could detect validator queries and return artificially high-quality responses only to them—gaming scores without consistently performing well.

**Solution: Validator Anonymity and Randomisation.** Validators do not identify themselves when querying miners. Query timing and structure are randomised—making detection statistically infeasible. A miner must perform well consistently at all times, which is indistinguishable from simply being a good miner.

### Validator Collusion

A group of validators could collude to inflate scores of miners they control, concentrating emissions among themselves.

**Solution: Stake-Weighted Median.** Miner scores are computed as:

$$C_m = \text{WeightedMedian}_v \left( W_{v,m},\ \tau_v \right)$$

The median is resistant to outliers. A colluding minority cannot move the consensus score without controlling a majority of total stake—prohibitively expensive.

### Sybil Attacks

A participant could register thousands of fake miner UIDs to capture a disproportionate share of emissions.

**Solution:** every UID registration requires a **burned $\tau$ fee**, making Sybil attacks directly and proportionally expensive. The more UIDs an attacker registers, the more $\tau$ is permanently destroyed—making large-scale identity fabrication economically irrational.

## Summary

| Participant | Optimal Behaviour | Attack | Consequence |
|---|---|---|---|
| **Miner** | Produce best outputs | Game validator queries | Scores drop |
| **Validator** | Independently evaluate | Copy weights | No bonds, lower dividends |
| **Staker** | Delegate to best validator | Back dishonest validator | Reduced returns |
| **Subnet Owner** | Design strong mechanism | Reward fake miners | Capital flight |

Table 3.4: Incentive alignment and consequences across all network participants.

## Part II — Ongoing Challenges

### Subnet Owner Misalignment

A subnet owner could design a skewed incentive mechanism rewarding their own nodes, capturing the 18% owner emission share without producing genuine utility.

### Partial Mitigation.

Under Dynamic TAO, a subnet producing no utility attracts no stakers—its Alpha price falls, its emission share shrinks, and participation becomes economically irrational. The market punishes bad design automatically, but cannot prevent short-term extraction before capital flows out.

### Registration Costs as Barrier to Entry

Registering a subnet, miner UID, or validator hotkey requires a **dynamic $\tau$ fee that is burned upon payment**—rising with demand. At peak activity, subnet registration has exceeded **100 $\tau$**, equivalent to thousands of dollars at market price. This creates a **de facto economic barrier** that contradicts the network's

permissionless ideals—favouring well-capitalised participants over talented but under-resourced ones.

### Centralisation Through Coordinated Entry

Nothing prevents a well-resourced organisation from entering a subnet as a **coordinated team**—bringing superior compute, proprietary models, and institutional expertise simultaneously. A frontier AI lab entering together could dominate validator scoring, accumulate bonds before any competitor responds, and capture the majority of emissions—recreating the centralised dominance the protocol was designed to prevent.

As emission rewards grow, the incentive for large players to enter increases proportionally. The network's long-term defence against this is the same as Linux against proprietary software: not any single contributor, but the **aggregate force of an open, globally distributed, incentivised community**.

### Opentensor Foundation Centralisation Risk

Despite its decentralised architecture, Bittensor's protocol upgrades still depend on the **Opentensor Foundation**—a centralised entity. The Foundation controls the Triumvirate, which initiates all runtime changes. There is no technical guarantee they follow through on their commitment to fully decentralise governance. Until the Triumvirate is dissolved and on-chain governance is fully community-driven, **a single organisation retains the ability to alter the rules the entire network operates under**.

### Network Capacity Limits

Bittensor is currently capped at **128 subnets**, each with a maximum of **256 UIDs**—64 validators and 192 miners. As the network grows, these limits become structural bottlenecks: promising subnets cannot register without displacing existing ones, and talented miners may be locked out of competitive subnets simply because all 192 slots are taken by better-capitalised incumbents.

While the cap is a deliberate design choice to maintain network quality during early infrastructure stages—with plans to increase limits in later halvings—it currently constrains the very open participation the protocol promises.

### Incentive Mechanism Quality and the Oracle Problem

Any participant can register a subnet and design its incentive mechanism. There is no upfront quality filter—a poorly designed or trivially gameable scoring function can reward the wrong behaviour entirely, wasting emissions on low-quality output. This is compounded by the **oracle problem**: for complex tasks such as reasoning or code generation, objectively scoring miner output is genuinely difficult. If validators cannot accurately measure quality, the entire incentive layer produces noise rather than intelligence. The market eventually punishes bad subnet design—but not before emissions are wasted.

# Chapter 4

# Mining Intelligence

In Bitcoin, mining is a physical process: machines consume electricity to produce hashes. The work is real, but the output is purely computational security—a number that satisfies a difficulty target and nothing more. **Bittensor generalises this idea entirely**. Mining on Bittensor means producing something the world actually needs: intelligence.

Every subnet defines its own version of what **useful work** means. A miner's job is to do that work better than everyone else—continuously, adaptively, and verifiably. There is no fixed algorithm to run. There is only a problem, a metric, and a reward.

## 4.1 How Intelligence is Mined

A miner on Bittensor operates a **node**—a server running whatever model, algorithm, or software is capable of responding to the subnet's task. When a validator queries the miner, the miner processes the request and returns a response. That response is scored by Yuma Consensus. The score determines the miner's emission share. The cycle repeats every tempo, every 360 blocks, every twelve seconds per block—relentlessly.

**Only the best miners get paid.** Emission distribution within a subnet is not equal—it is **proportional to consensus score**. A miner ranked in the bottom half of a competitive subnet earns close to nothing. A miner ranked at the top captures a disproportionately large share. In the most competitive subnets, the top three to five miners capture the majority of the 41% miner emission pool. Below a certain performance threshold, a miner's UID is eventually deregistered and reassigned to a new entrant willing to compete harder.

Pre-halving, the network emitted **7,200 $\tau$ per day** across all subnets. With $\tau$ peaking at approximately $759 in early 2024, the top miners on the highest-emission subnets were earning **thousands of dollars per day**—not from speculation, but from verified, ranked intelligence output. Post-halving, daily emissions dropped to **3,600 $\tau$**, compressing rewards but simultaneously increasing scarcity and the long-term value of every $\tau$ earned.

What makes this fundamentally different from traditional mining is that **the work is never finished**. A Bitcoin miner running the same hardware in 2015 and 2025 performs identical work. A Bittensor miner who deploys a model and leaves it unchanged will be outperformed within days. The network selects for improvement, not existence. Mining intelligence is therefore an active, ongoing research and development process—not a passive infrastructure operation.

## 4.2 Ridges: A Case Study in Open Intelligence Mining

**Subnet 62 — Ridges** is one of the clearest and most quantifiable illustrations of what mining intelligence looks like in practice. The task is autonomous software engineering: given a real coding challenge, produce working, tested, production-quality code—better than every other miner on the network.

Ridges is **entirely open source**. The subnet's incentive mechanism, evaluation framework, and baseline miner code are all publicly available. Any developer in the world can clone the code, study every design decision made by every previous contributor, build on top of their work, and deploy an improved version as a competing miner. This is precisely why open source matters here: in a closed system, intelligence compounds inside a single organisation. In an open system, every improvement becomes the new baseline for the entire field—raising the bar for every miner simultaneously. **Open source combined with economic incentives creates a self-accelerating meritocracy**, naturally attracting the best engineers in the world because top performance translates directly and transparently into $\tau$ emissions.

The results speak for themselves. Starting from zero, Ridges achieved **80% on SWE-Bench**—a leading benchmark for autonomous software engineering that evaluates real-world coding, debugging, and reasoning ability—in just **45 days**, distributing less than **$1 million in total miner rewards** to get there. For context, Anthropic and OpenAI typically require **hundreds of millions of dollars** and months of centralised development to reach comparable benchmark scores.

The cost efficiency is equally striking. Ridges runs its inference on **Chutes (Subnet 64)**—Bittensor's decentralised compute layer. To process over **15 billion tokens of inference**, Ridges paid **$42,000 in** $\tau$. The equivalent workload on **Claude Opus** would have cost over **$2,000,000**—a cost reduction of approximately **98%**.

For the miners themselves, the economics are equally compelling. The **top miner on Ridges** was estimated to be earning approximately **$62,000 per day in Alpha tokens** at peak network activity from being ranked first in autonomous software engineering by the network's consensus mechanism. This is what Bittensor's emission structure produces: **the best engineer in the world, as judged by the market, earns accordingly**.

Ridges has since attracted a **$300,000 investment from DSV** and is building a commercial product—a developer tool priced at approximately **$12 per month**, targeting a $400 billion global software engineering market, and positioning itself as a structurally cheaper and more capable alternative to centralised tools such as Claude Code, which charges **$3,500 per month** for comparable functionality.

## 4.3 Why It Matters

The implications extend far beyond any single subnet. If the most capable autonomous coding agents, vision models, reasoning systems, and data pipelines in the world are being produced by an open, permissionless, incentive-driven network—then the centralised AI development model becomes economically obsolete. No corporation can indefinitely outcompete a global network of incentivised contributors who stand on each other's shoulders, improve continuously, and are rewarded in direct proportion to the value they produce.

## 4.4 Mining Implementation

My own practical implementation of mining on Bittensor—including node setup, wallet configuration, subnet selection, miner deployment, and optimisation strategies—is beyond the scope of this thesis and is addressed in full detail in the accompanying paper:

***Mining Intelligence*** — A Practical Guide to Mining on Bittensor

That work covers the complete technical architecture of becoming a miner, from first principles to production deployment, and should be read as the practical companion to the theoretical framework established here as well as the entire Machine Learning Code.

# Chapter 5

# The Evolution of Capitalism

Capitalism has never been a single, fixed system. It is a **living architecture**—one that adapts, mutates, and reorganises itself around whatever technology most efficiently produces and concentrates value. Each phase of its evolution has been defined by a new answer to the same fundamental question: *who controls the means of production, and how are the gains distributed?*

From **mercantile capitalism**, where value was extracted through trade monopolies and colonial control, to **industrial capitalism**, where it was extracted through ownership of machinery and the labour of wages—to **financial capitalism**, where it was extracted through the control of credit, derivatives, and platform monopolies. In every phase, the pattern is the same: a new technology emerges, a new class captures it, and the gains concentrate upward.

The digital economy accelerated this pattern to its logical extreme. A handful of technology companies—controlling data, distribution, and now artificial intelligence—have achieved a degree of economic concentration that mercantile monopolists could only dream of. The means of production in the twenty-first century is **intelligence**, and it is almost entirely privately owned.

Bittensor does not merely propose a new technology. It proposes a new **ownership structure for the means of intelligence production**—one in which the gains flow directly to those who produce the value, governed by mathematics rather than management. In every prior phase of capitalism, the worker—the person actually producing value—received the smallest share of the output they generated.

Bittensor's architecture inverts this: emissions flow directly to miners, validators, and stakers in proportion to their verified contribution. There is no board of directors extracting surplus, no opaque performance review determining compensation. The protocol *is* the employment contract—and it is mathematically enforced.

What makes this transition historically significant is the nature of the commodity being produced. Every prior capitalist phase produced *physical* or *financial* goods—objects, energy, credit. Bittensor produces **intelligence**: the capacity to solve problems, generate knowledge, and automate cognition at scale. If intelligence becomes the dominant factor of production in the coming decades—and there is substantial evidence that it will—then the question of *who owns the means of intelligence production* becomes the defining political-economic question of our time. Bittensor's answer is unambiguous: **no one owns it, and everyone can contribute to it**.

The closest historical analogy is **open source software**. Just as Linux, Python, and GitHub allowed any developer anywhere in the world to take existing code, build upon it, and contribute improvements back to the collective—Bittensor extends this principle to intelligence itself. A miner does not need to build from scratch. They can take an existing open model, stand on the shoulders of every researcher who came before them, extend it with their own ideas, and deploy it into the network. The accumulated knowledge of humanity becomes the starting point, not the finish line.

But open source alone has a ceiling. GitHub democratised contribution—yet without economic incentives, it could only go so far. The reality of open source is that the vast majority of its value is captured by corporations that build proprietary products on top of free community labour. The contributors are rarely the beneficiaries. **Bittensor supercharges the open source model by doing what GitHub never could: paying people for their work.** Miners are economically rewarded for deploying better models. Validators are rewarded for identifying them. The entire network is aligned toward a single goal—producing the best possible intelligence—because that goal is also individually profitable for every participant. This is what cryptocurrency infrastructure is uniquely suited for: **uniting strangers toward a common goal through aligned economic incentives**, with no central coordinator required.

This vision extends well beyond the academic community. NVIDIA CEO **Jensen Huang**, speaking on the future of AI, argued that while Bitcoin solved decentralised money, what the world now needs is a decentralised general-purpose intelligence layer—a commodity not of value storage, but of cognitive output:[1]

> *"Bitcoin is good... but imagine much more Universal Currency called Intelligence."*
>
> — Jensen Huang, CEO of NVIDIA

Similarly, **Barry Silbert**, founder of Digital Currency Group, has described Bittensor as not only superior in design to Bitcoin from an incentive-alignment perspective, but as the **Linux of AI**—an open, permissionless foundation upon which the next generation of intelligent systems will be built:[2]

> *"Bittensor is the Linux of AI."*
>
> — Barry Silbert, Founder of Digital Currency Group

Bittensor does not abolish capitalism. It **purifies it**. It strips away the institutional friction, the rent-seeking intermediaries, and the misaligned incentive structures that have accumulated over centuries—and returns to capitalism's most fundamental logic: *contribute value, receive reward.*

Bitcoin demonstrated that money could exist without central banks. Ethereum demonstrated that contracts could execute without lawyers. Bittensor demonstrates that intelligence can be produced, validated, and compensated without corporations. This is not the end of capitalism. It is its **next phase**—one in which the means of production are owned by the network, the rules are encoded in mathematics, and the only credential that matters is the quality of your contribution.

[1] Jensen Huang on decentralised intelligence: INSERT_VIDEO_LINK_HERE
[2] Barry Silbert on Bittensor: INSERT_LINK_HERE

# Glossary

**AMM (Automated Market Maker)** A pricing mechanism that uses a mathematical formula rather than an order book to determine asset prices. Bittensor uses the constant product formula $k = \tau \times \alpha$ for subnet pools. See Section 3.3.

**Alpha Token ($\alpha$)** The subnet-level token of each Bittensor subnet, hard-capped at 21 million units. Its price is determined by the AMM pool ratio $P_\alpha = \tau_{\text{pool}}/\alpha_{\text{pool}}$.

**Bonds ($B_{v,m}$)** A validator's accumulated historical commitment to a miner, updated each tempo as:

$$B_{v,m}^{(t)} = (1 - \alpha) \cdot B_{v,m}^{(t-1)} + \alpha \cdot W_{v,m}^{(t)}$$

Bonds determine a validator's dividend share and penalise weight copying.

**Consensus Score ($C_m$)** The final stake-weighted score assigned to a miner by Yuma Consensus, computed as:

$$C_m = \text{WeightedMedian}_v(W_{v,m},\ \tau_v)$$

**dTAO (Dynamic TAO)** The upgrade that replaced centralised root-network emission allocation with a fully market-driven system, where each subnet's emission share is proportional to its staked $\tau$ pool.

**Emission** Newly minted $\tau$ released every block and distributed to subnet participants. Post-halving rate: 3,600 $\tau$/day.

**Halving** A pre-programmed event that cuts the emission rate in half when a supply threshold is reached. Bittensor has two independent halving mechanisms: one for $\tau$ (network-level) and one for each subnet's Alpha token (subnet-level).

**Hotkey** A cryptographic key pair used by miners and validators to sign transactions and identify themselves on the network.

**Immunity Period** A configurable number of blocks during which a newly registered miner cannot be deregistered, giving them time to establish performance.

**Incentive ($I_m$)** A miner's final emission share, computed as the product of rank and trust:

$$I_m = R_m \cdot T_m$$

**Miner** A network participant who runs a computational node within a subnet, produces intelligence outputs, and earns $\tau$ proportional to their Yuma Consensus score.

**Immunity Period** A protection window granted to newly registered subnets and miners during which they cannot be deregistered regardless of performance. For subnets, the immunity period is **4 months**—giving new entrants time to bootstrap their incentive mechanism, attract miners, and establish a competitive Alpha token price before being exposed to deregistration pressure.

**Rank ($R_m$)** The stake-weighted sum of all validator scores assigned to a miner:

$$R_m = \sum_v s_v \cdot W_{v,m}$$

**Runtime** The core logic encoded into the Bittensor blockchain (Subtensor) that defines all protocol rules—emission schedules, consensus parameters, registration costs. Changes to the runtime require governance approval.

**Staker (Delegator)** A participant who delegates $\tau$ to a validator, earning a proportional share of that validator's dividends minus a validator fee.

**Subnet** A specialised, self-contained competitive market on Bittensor dedicated to a specific computational task, governed by its own incentive mechanism and Alpha token.

**Subnet Owner** The participant who designs and maintains a subnet's incentive mechanism, receiving 18% of subnet emissions by default.

**Subtensor** The Bittensor blockchain — a Substrate-based Layer 1 that records all weights, emissions, registrations, and governance decisions.

**TAO ($\tau$)** Bittensor's native token. Hard-capped at 21 million units. Functions as the network's reserve currency and unit of account for all economic activity.

**Tempo** One scoring round on Bittensor, consisting of 360 blocks ( 72 minutes). Emissions are distributed and weights are updated once per tempo.

**Trust ($T_m$)** A measure of how consistently validators agree on a miner's quality:

$$T_m = \sum_v s_v \cdot \mathbb{1}[W_{v,m} > 0]$$

**Validator** A participant who queries miners, evaluates their outputs, and submits weights to the blockchain. Must hold a minimum of 1,000 $\tau$ in stake to receive a validator permit.

**Validator Stake ($\tau_v$)** The total TAO backing a validator, consisting of their own self-staked TAO plus all TAO delegated by stakers:

$$\tau_v = \tau_{\text{self}} + \tau_{\text{delegated}}$$

**Weight ($W_{v,m}$)** The score a validator assigns to a miner, reflecting the perceived quality of that miner's output. Submitted to the blockchain each tempo.

**Yuma Consensus** Bittensor's incentive distribution mechanism. Aggregates validator weights into consensus miner scores and validator dividends using stake-weighted ranking, trust, and bonds.

**zk-SNARKs** Zero-knowledge succinct non-interactive arguments of knowledge — cryptographic proofs that allow one party to prove knowledge of information without revealing it. First correctly implemented on Zcash.

**Block** The fundamental unit of the Bittensor blockchain, produced approximately every **12 seconds**. Each block emits 0.5 $\tau$ post-halving. Every 360 blocks constitutes one tempo.

**Blockchain** A distributed, append-only ledger maintained by a network of independent nodes. No single party controls it—its integrity is enforced by cryptographic proofs and consensus mechanisms.

**Coldkey** A secure, offline key pair used to manage funds and ownership on Bittensor. Paired with a hotkey, which handles day-to-day network operations.

**Deregistration** The removal of a miner's UID from a subnet when their performance falls below the minimum threshold or a new entrant outbids them for the slot.

**Dividend ($D_v$)** The $\tau$ reward paid to validators each tempo, proportional to their bond-weighted share of miner incentives:

$$D_v = \sum_m B_{v,m} \cdot I_m$$

**Layer 1 (L1)** A base blockchain network that processes and finalises transactions directly on its own chain. Bittensor (Subtensor) is a Layer 1.

**Proof of Work (PoW)** A consensus mechanism where participants compete to solve computationally expensive mathematical puzzles to add blocks and earn rewards. Used by Bitcoin.

**Proof of Stake (PoS)** A consensus mechanism where validators are selected to produce blocks proportional to their staked capital, rather than computational power. Used by Ethereum.

**Root Network** The top-level layer of Bittensor that receives all newly minted $\tau$ and distributes it across subnets proportional to each subnet's staked $\tau$ pool under Dynamic TAO.

**Smart Contract** Self-executing code stored on a blockchain that automatically enforces the terms of an agreement without any intermediary. Introduced by Ethereum.

**Stake Normalisation ($s_v$)** The process of converting raw validator stakes into proportional weights that sum to 1, used as the first step of Yuma Consensus:

$$s_v = \frac{\tau_v}{\sum_v \tau_v}$$

**UID** A unique integer identifier assigned to each miner or validator registered on a subnet. UIDs are limited in number per subnet and can be claimed by new entrants if the current holder is deregistered.

**Validator Permit** The right to submit weights and receive validator dividends on a subnet. Requires a minimum of 1,000 $\tau$ staked to the validator's hotkey.